# Capstone 6 – Mini-Guide: Reading Beginner Python Safely

A one-page companion for learners who use AI to explain code. Every point below was checked by running code.

## Rule 1: Run it before you trust it

An AI explanation is a claim, not a fact. Paste the code into Google Colab, run it, and compare the real output with what the AI said.

## Rule 2: Know the five common traps

| Trap | Example | What to remember |
|---|---|---|
| End values are excluded | `range(1, 6)` → 1 to 5 | Same for slices: `fruits[1:3]` gives indexes 1 and 2 |
| Lists change as you edit | `insert(1, x)` shifts later items right | Re-trace the list after every change |
| File modes matter | `"w"` overwrites, `"a"` appends, `"r"` reads | Check the mode before trusting any file explanation |
| Default arguments are shared | `def f(x, a=[])` | Use `a=None` and build the list inside |
| Empty loops still work | `range(2, 1)` is empty, so `factorial(0)` stays 1 | Ask "what if the input is 0 or empty?" |

## Rule 3: Quick concept card

- **Loop (`for`)**: repeat for each item in a sequence.
- **Loop (`while`)**: repeat while a condition is True. Always change something inside the loop.
- **Function**: `def name(parameters):` … `return value`.
- **List**: ordered, changeable, indexes start at 0, `-1` is the last item.
- **Dictionary**: key → value pairs. `d.get(key, 0)` avoids errors for new keys.
- **File**: always use `with open(name, mode) as f:` so it closes by itself.

- **String reversal**: `text[::-1]`.

## Rule 4: Spot outdated or unsafe suggestions

| Red flag | Why | Use instead |
|---|---|---|
| `print "text"` | Python 2 | `print("text")` |
| `raw_input`, `xrange` | Python 2 | `input`, `range` |
| `eval(input())` | Runs any typed text as code | `int(input())` |
| `os.system(...)` with user input | Can run harmful commands | Avoid |
| `open()` without `with` | File may stay open | `with open()` |

## Rule 5: Ask better questions

1. "Explain line by line for a complete beginner."
2. "Show a table of each variable's value after every loop pass."
3. "What would happen if the input were empty or 0?"
4. "Give 3 practice problems with answers hidden."
5. Then run everything yourself.

## Rule 6: Practise with the checker

Use the `check_answer()` cell from the toolkit. Write your function, run the cell, and read PASS/FAIL for each test case before looking at any AI solution.

## Self-check

After reading this guide, take the 10-question quiz (document 5).